



A review of cavability evaluation in longwall top coal caving



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ABSTRACT

Longwall Top Coal Caving has been considered as one of the most effective technologies for the extraction of underground thick coal seams. A large number of studies on the applicability of Longwall Top Coal Caving into new mine sites have linked the success of its application to the geomechanical understanding of the cavability of the top coal. The paper aims to improve the knowledge of the top coal cavability evaluation. A range of parameters that affect the top coal cavability were first identified. Afterward, a number of cavability assessment methods and classifications were reviewed. The result is important in that it assists researchers in developing an advanced and reliable tool for the top coal cavability evaluation.

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1. Introduction

The underground extraction of the thick coal seam with thickness in excess of 4.5 m is faced with operational, technical, safety and economic problems. Several longwall mining methods such as Multi-Slice Longwall (MSL), High Reach Single Pass Longwall (SPL) and Longwall Top Coal Caving (LTCC) have been mainly applied to extract thick coal seams. Multi-Slice Longwall method divides a thick seam into separate slices that are extracted by a conventional longwall method. The method takes advantage of conventional longwall equipment; however, it suffers major disadvantages including operational costs and ground control issues. High Reach Single Pass Longwall is a conventional longwall method with a larger working height. The method has great benefit in the 4.5–5.0 m mining height range. When mining height exceeds 5.0 m, due to limitations such as equipment design and face conditions, this method has struggled to optimise productivity and to control face stability. Compared with other methods, LTCC offers a method that can extract up to 80% of the seam in the 5–9 m thickness range. The development cost per tonne is reported to be significantly reduced. Compared to SPL, LTCC results in improved face control, smaller and less expensive equipment, and improved spontaneous combustion control [1,2]. The method, however, is not applicable for all thick coal seams. Coal seam cavability, roof strata characteristics and high horizontal stress are

examples of mining conditions that must be considered to attain a safe and effective LTCC operation [3].

Longwall Top Coal Caving was initially developed in the former Soviet Union and France in the 1950s and 1960s. It was then applied in the former Yugoslavia, Hungary, Romania, the former Czechoslovakia and Turkey [4–6]. The ‘Soutirage’ method which was developed in France in the 1960s is considered as the original form of LTCC [7]. The system includes a conventional longwall face operating at the base of a thick coal seam and the upper part of seam is extracted by caving (Fig. 1). This method faced major issues including dust and significant face disruption due to the top coal drawing [8]. After the mid-1980s, LTCC was abandoned in Europe. The major reason was that the levels of productivity from LTCC faces were less than that from conventional longwalls during that time. Since the late 1980s, LTCC has been introduced, developed and improved in China. The method is widely applied to extract thick seams in China with a significant innovation on equipment. The coal draw window was relocated to the rear canopy instead of being at the roof canopy while a second Armoured Flexible Conveyor (AFC) was designed behind the support base (Fig. 2). The improved designs of support and AFC and the additional extraction of top coal are the major differences between LTCC and a conventional longwall method. A conceptual model of LTCC is shown in Fig. 3. Recently, the Chinese mining industry has successfully developed LTCC equipment for the extraction of the 14–20 m ultra-thick seams with a cutting height up to 5 m [9]. The LTCC method for mining thick seams with large dip angles up to 41° has also been introduced in China with the top coal recovery rate up to 80% [10]. Apart from China, LTCC has been employed in

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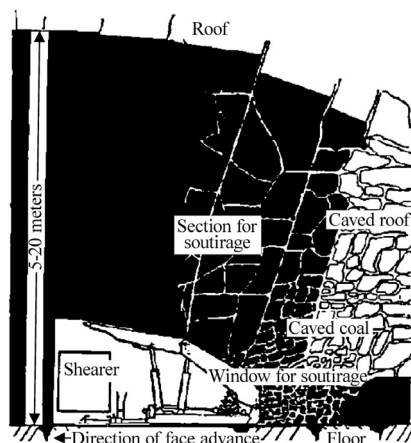


Fig. 1. 'Soutirage' mining method (after Schneiderman [7]).

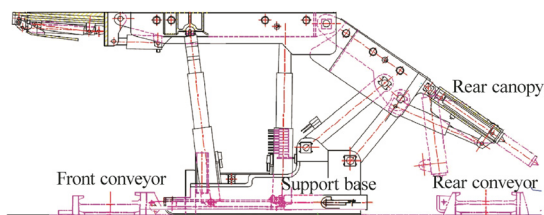


Fig. 2. LTCC support with double conveyors (after Xie and Zhao [13]).

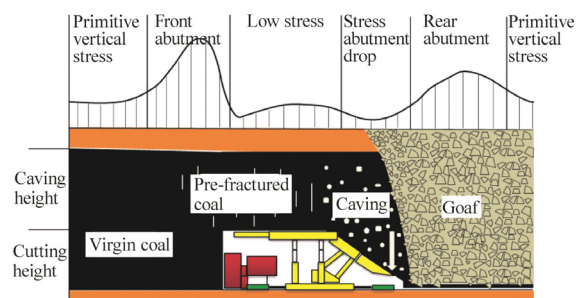


Fig. 3. LTCC conceptual model (after Xu [5]).

Australia and Vietnam [2,11]. Efforts are being made to introduce this method into India [12].

The successful application of LTCC in China in terms of productivity, cost and safety has promoted its application on a world scale. The variety of mining conditions in different countries requires the development of an engineering tool that can evaluate the LTCC applicability in various mine sites. The top coal cavability assessment has been the interest of LTCC studies. Empirical and observational methods have been widely applied to evaluate cavability in Chinese LTCC practices [5]. Various numerical modelling methods have been used to study caving mechanisms [6,13–18].

Physical models have been designed to understand the impact of overburden strata on cavability or to study the top coal drawing [17,19,20]. Analytical solutions have also been utilised to analyse cavability or face stability [5,21–23].

A number of cavability evaluation methods have been developed [5,11,16,21,24]. However, there is no single method that can be universally applied. There is a need for an advanced and reliable assessment tool for various mine site conditions. A systematic understanding of the cavability controlling parameters and of the evaluation approaches is a necessary precursor to the development of any advanced cavability assessment method.

2. Parameters influencing top coal cavability

In a significant number of cavability studies, the evaluation systems are commonly derived from an understanding of the parameters that seem to have major impacts on caving performance. This section aims to analyse the effect of geological and geotechnical factors on the top coal cavability. These parameters can be categorised into four groups including coal seam characteristics, surrounding rock strata characteristics, stress conditions and others.

2.1. Coal seam characteristics

The coal seam characteristics which have been investigated in cavability studies mainly include the coal strength, coal discontinuities, coal cutting height and top coal thickness. The top coal cavability is an inherent characteristic that indicates the sensitivity and possibility of coal to cave under the action of strata abutment stress. The coal strength which indicates its resistant ability to failure directly affects the material failure and therefore greatly influences cavability [5]. The Uniaxial Compressive Strength (UCS) is generally used to represent the rock strength. The importance of coal UCS compared to other parameters on overall cavability rating in several studies is shown in Table 1 [5,11,25]. The weighting values show that the top coal cavability is highly sensitive to the coal strength. It is noted that UCS values are normally representative of material strength in laboratory scale. Thus, the value should be scaled by a certain factor to represent the material strength in field scale. The techniques for the determination of rock and coal strength can be found in literature [26–29]. The approaches for scaling material strength from laboratory to field scale were presented in many studies [30–34].

Discontinuity is commonly employed as a collective term for all fractures or features in a rock mass that have zero or low tensile strength [27]. In most rock masses, discontinuities form planes of weakness or surfaces of separation and generally control the mechanical behaviour of rock mass [36]. The characterisation of discontinuities can be found in Esterhuizen [37], Laubach et al. [38] and Brady and Brown [27]. The effect of discontinuities on roof rock stability has been incorporated in the widely known rock mass classifications [39–43]. In this paper, discontinuities are limited to bedding planes, joints and face cleats. Due to the stratified nature of coal measure rocks, the high discontinuity density in coal seam significantly affects cavability. Some examples for the importance of coal discontinuities compared to other parameters on

Table 1
Weightings of coal strength on overall cavability rating (after Xu [5], Dao [11], Vakili [25]).

Author	Method	Parameter	Weighting (%)
Jia [35]	Fuzzy cluster	UCS	38.2
Shandong coal mining [5]	Statistical analysis	Cover depth/coal UCS ratio	23.0
Vakili [25]	Top coal recovery rating	UCS	42.0
Dao [11]	Caving performance index	UCS	13.1

Table 2

Weightings of coal discontinuities on overall cavability rating (after Xu [5], Dao [11], Vakili [25]).

Author	Method	Parameter	Weighting (%)
Jia [35]	Fuzzy cluster	Joint spacing	36.3
Shandong Coal Mining [5]	Statistical analysis	Cleat spacing	14.0
Vakili [25]	Top coal recovery rating	Vertical joint spacing	35.0
Dao [11]	Caving performance index	Vertical joint spacing	10.4
		Bedding spacing	34.4

overall cavability rating are presented in Table 2 [5,11,25]. It is noted that previous LTCC studies mostly focused on the discontinuity spacing.

The other important characteristics such as discontinuity persistence and spatial orientation have not been studied in detail. The reason might come from the difficulty in the collection of these characteristics in reality. The cutting height and top coal thickness are considered to affect the quantity and quality of coal caving. If the cutting height is kept unchanged and the top coal becomes thicker, the void space may not be enough to allow the complete rotation and movement of top coal. This leads to a poor top coal fragmentation and less cavability [25]. Dao stated that the resistance of top coal to failure depends on the coal strength and its thickness [11]. If the coal strength is the same, a thinner top coal is more likely to fail compared to a thicker top coal. According to Xu [5], the ratio of top coal thickness to cutting height should be reasonably set to obtain a successful caving. The ratio values generally vary from one to three and may reach to five for Chinese LTCC mines. In the application of LTCC into ultra-thick seams, this ratio has more influence on the top coal recovery and requires to be further investigated [44]. Examples for the importance of coal thickness compared to other parameters on overall cavability rating are presented in Table 3. It is noted that the band thickness in top coal partly contributes to caving [5]; however, its influence is less important and was not incorporated in many cavability classifications [11,16,21,24].

2.2. Surrounding rock strata characteristics

The characteristics of surrounding rock strata are considered to contribute to the success of top coal caving. One of the basic conditions for a LTCC mining is that the immediate roof must be of certain thickness and caves immediately or with little delay after the advance of support [5]. If an immediate roof is weak and thin, it will converge to top coal due to the load from upper strata and its own weight. This causes an increase of stress on top coal that facilitates the coal failure. Therefore, top coal will be easier to cave. Conversely, if an immediate roof is strong and thick, it can resist the load transference from upper strata and its own weight on top coal. As a result, top coal will be more difficult to cave. The main roof affects the stability of immediate roof and thus influences the top coal behaviour. If a main roof is thinly bedded and weak, the bedding layers can cave simultaneously with the immediate roof. This causes an additional load from broken main roof

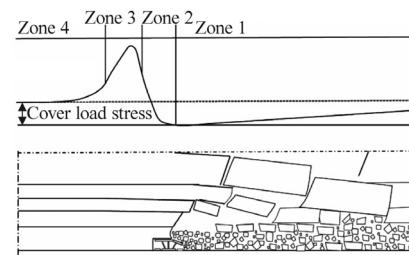
rocks on top coal. If a main roof is thickly bedded or massive, weighting events with long intervals can occur. In that case, the strata will inhibit the transference of vertical stress on top coal and consequently, a less cavability is anticipated [11]. In addition, a weak floor rock may result in support instability and/or floor heaving. These issues disturb caving sequence and thus affect cavability indirectly.

The significance of overlying strata on top coal behaviour has been evaluated in few studies [1,5,24,25]. However, they mostly focused on the strength of roof rocks while neglected the characteristics of bedding layers and strata thickness. This limitation probably comes from the difficulty in identifying roof structures in sedimentary rock that often consists of various layers with different engineering characteristics [45]. The determination of the extent of immediate and main roof strata can be found in Kidybinski and Babcock, Peng and Chiang, Whittaker et al. and Majdi et al. [46–49].

2.3. Stress conditions

Prior to an underground excavation, the rock medium is subjected to an initially equilibrium state of stress. The underground mining causes the stress redistribution and forms a new equilibrium stress state. The mining-induced stresses directly affect the material failure and therefore influence the top coal cavability. The determination of pre-mining stress can be referred to numerous studies [27,29,50–52].

For LTCC mining, the vertical stress component is redistributed and results in different stress zones (Fig. 4). The formation of high vertical abutment stress zones induces the top coal to fail and to cave under the impact of gravity. The horizontal stress component, however, has an ambiguous effect on caving [11,25]. On one hand,

**Fig. 4.** Vertical stress distribution at a LTCC face cross-section (after Dao [11]).**Table 3**

Weightings of coal thickness on overall cavability rating (after Xu [5], Dao [11], Vakili [25]).

Author	Method	Parameter	Weighting (%)
Jia [35]	Fuzzy cluster	Cave/cut ratio	7.2
Shandong coal mining [5]	Statistical analysis	Cut/cave ratio	14.0
Vakili [25]	Top coal recovery rating	Top coal thickness	19.0
Dao [11]	Caving performance index	Seam thickness	31.8

the horizontal stress is a cause of cutter roof failure. On the other hand, it can constrain the load transference from upper to lower strata. This reduces the level of coal failure and finally decreases cavability. The stress redistribution induced by LTCC mining has been investigated in few studies [6,14,15]. It is noted that the stress magnitudes are difficult to measure due to the cost and time restraint [53]. This may lead to the limitation of stress data for a comprehensive cavability study.

2.4. Other parameters

Apart from the major parameters discussed, others such as panel design, seam dip and ground water may need to be considered in the evaluation of cavability. The effect of panel orientation with respect to vertical joints and to major horizontal stress direction on caving was studied by Vakili [25]. The study concluded that a face orientation at an angle of 45° to the face vertical joint might cause a better top coal recovery in the centre of panel width compared to the non-oriented panel. Meanwhile, the top coal loss near gate-ends was greater. The author also stated that a higher angle between roadways and the major principal stress can improve caving performance. It is noted that these results were preliminary and more in-depth studies are required.

The effect of seam dip on cavability was investigated by Dao [11]. He found that the dip angle played the least important role on caving compared to other parameters. The ground water is widely known to change rock mass strength and therefore affects the caving behaviour. However, considering the simplicity of input data collection, many LTCC studies have not taken into account the ground water effect in their cavability assessment methods. Future studies may need to deal with such issues in order to establish a more comprehensive evaluation.

3. Top coal cavability evaluation and classification

The evaluation and classification of rock mass stability is of great importance for the ground control design. A number of rock mass classifications have been developed for various applications in mining and civil engineering practice such as Q index [39], Rock Mass Rating [40], Mining Rock Mass Rating [41] and Stability Number [42]. These classifications were mainly designed for a wide range of rock mass strength. Therefore, their sensitivity on soft rocks such as coal is very small [25].

For the stability classification of coal measure rocks, Molinda and Mark developed the Coal Mine Roof Rating (CMRR) system [43]. However, the classification was initially designed to evaluate the stability of roof rock rather than coal seam. It is thus less sensitive to different categories of coal properties [25]. For the evaluation of roof stability in conventional longwalls, Medhurst et al. developed a caving chart for Australian longwall operations (Fig. 5) [54]. The chart was designed based on support pressure,

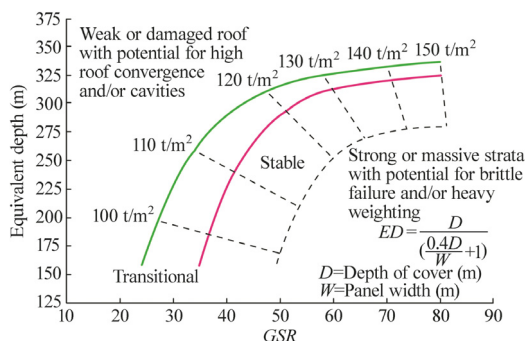


Fig. 5. Caving response for Australian longwall operations (after Medhurst et al. [54]).

panel layout and the strata characterisation. The strata were characterised using geophysical logs. Similar to CMRR, the chart was not developed for the instability assessment of top coal in the roof. Its applicability in LTCC may require future investigation. Other approaches for caving assessment in conventional longwall were reviewed in detail by Singh [55].

The characteristics of coal measure rocks and the additional extraction of top coal in LTCC require the development of a cavability evaluation system that is different from those applied for either non-coal measure rocks or traditional longwall mining. The following reviews of the well-known top coal cavability evaluations aim to provide readers with an insight into the detailed approaches which have been adopted.

3.1. Chinese approach

With more than 30 years of LTCC application, the Chinese coal industry has used several cavability assessment methods. The following methods were cited by Xu and Cai et al. [5,56].

3.1.1. Fracture index method

Zhang and Wang developed a cavability assessment method based on a theoretical analysis [57]. They stated that the formation of plastic zone ahead of face line is the prime condition for top coal to cave. Based on this statement, a cavability classification was proposed as shown in Table 4 where r is the cover rock density, kg/m³; g is the gravity, m/s²; H is the cover depth, m; R_c is the uniaxial compressive strength of coal, MPa; and k_m is the peak front abutment stress, MPa.

The method introduced a cavability assessment criterion with regard to the top coal fracturing. Parameters including coal strength, cover depth and peak abutment stress were incorporated in the cavability index. It is seen that the effect on caving of other coal seam characteristics such as thickness and structures was not studied. This significantly limits the accuracy of the cavability assessment.

3.1.2. Fuzzy cluster evaluation method

The fuzzy cluster evaluation method was empirically developed by Jia [35]. The input parameters from a range of Chinese LTCC faces and the cavability classification are illustrated in Table 5. The classification rating from 1 to 4 means excellent cavability, good cavability, medium cavability and poor cavability respectively. The combination of the parameters known from previous caving studies with the new factors such as coal structure or overlying strata makes this method more comprehensive compared to previous assessments. The database of caving performance from practical LTCC faces provides a useful source for the validation in other cavability classifications. However, guidelines for the categorisation of joint/cleat, the categorisation of main roof, the derivation of filling factor and the classification of cavability are not available. The application of this method may be difficult for sites whose mining conditions are not similar to those in Table 5.

Table 4
Cavability classification in fracture index method (after Zhang and Wang [57]; cited by Xu [5]).

Characteristic	Classification
$r \times g \times \frac{H}{R_c} < \frac{1}{k_m}$	Cavability is very poor. Top coal is very difficult to cave naturally
$\frac{1}{k_m} < r \times g \times \frac{H}{R_c} < 0.5$	Top coal can cave but external fracturing techniques are required
$0.5 < r \times g \times \frac{H}{R_c}$	Caving is easy by natural abutment pressure

Table 5

Cavability classification in fuzzy cluster method (after Jia [35]; cited by Xu [5]).

Seam	Coal UCS (MPa)	Cover depth (m)	Joint-cleat category	Band thickness (m)	Filling factor by caved immediate roof	Main roof category	Cave to cut ratio	Cavability classification
Seam 3, Puhe Mine, Shengyang	15–20	150	1	0.12	1.543	2	5.220	1
Seam 8, No 1 Mine, Pindingshan	20	480	3	0.15	1.167	2	1.571	2
Seam 15, No 1 Mine, Yanquan	30	250	2	0.20	0.325	2	1.540	2
Seam 15, Michung, Zhendzhou	5–10	130	1	0	1.143	1	2.360	1
Seam 3, Fenghuangshang Mine, Jincheng	38	210	4	0.40	1.600	3	1.400	3
Seam 4, No 4 Mine, Yanquan	30	240	2	0.20	0.312	2	1.140	2
Seam 3, Wangzhuang Mine, Luan	25	254	2	0.20	1.134	2	1.450	2
Seam 11, 12, No 2 Mine, Datong	32	350	5	0.30	0	4	1.679	4
Seam 5, Liujialiang Mine, Xuangang	7	230	1	0.25	0.500	1	1.400	1
Seam 2, Xingtai Mine, Xingtai	20	360	2	0.40	0.406	2	1.462	2
Seam 3, Xinlongzhuang Mine, Yanzhou	24	412	3	0.30	1.159	2	1.589	2
Seam 10, 11, Shuiyu Mine, Fengxi	5–11	210	2	0.31	1.460	2	2.674	1
Da Seam, No 6 Mine, Hebi	15		3	0.21	1.300	2	1.571	2
Seam 7, Shanghejing Mine, Xuzhou	30	150	2	0	0.525	2	2.800	2

Table 6

Weightings of parameters on cavability in statistical analysis method (after Xu [5]).

Factor	Cover depth/coal UCS ratio	Immediate roof	Main roof	Cut/cave ratio	Cleat space	Band thickness	Band UCS
Weighting A_i	0.23	0.12	0.10	0.14	0.14	0.12	0.15

Table 7

Cavability classification in statistical analysis method (after Xu [5]).

Cavability	Very good	Good	Medium	Poor	Very poor
Index M	0.90–1.00	0.80–0.90	0.65–0.80	0.50–0.65	<0.5

3.1.3. Statistical analysis method

The Shandong Coal Mining Administration developed another empirical method for the top coal cavability assessment. The first step of implementation is to rate seven affecting parameters which are shown in Table 6. The specific rating values of factors can be found in Xu [5]. The next step is to multiply these rating by their weightings A_i (Table 6). The cavability index M is calculated by Eq. (1).

$$M = \sum_{i=1}^7 A_i \times \mu_i \quad (1)$$

where A_i is the weighting of factor i ; μ_i is the rating of factor i . Finally, the index M is compared to cavability classification outlined in Table 7.

Compared to other Chinese cavability evaluations, this method seems to be the most comprehensive system due to the following advantages. Firstly, the method is easy to be applied. This results from the available guideline for the evaluation of each parameter and from the detailed procedure of the method implementation. Secondly, the method assesses a wide range of factors including the coal, band and overlying strata characteristics as well as the operational parameter. It is noted that the effect of stress regime on caving was not incorporated in the method. As discussed previously, this may reduce the reliability of assessment results.

3.2. Numerical modelling approach

Empirical, observational, physical modelling and numerical modelling approaches have been commonly applied to study the rock mechanics in longwall mining. The empirical approach may only benefit the cavability evaluation at new mine sites having similar conditions to those whose cavability had been assessed.

Due to cost and practicality, the field monitoring or measurement approach should be carefully designed prior to the application. For the physical modelling, it should not be solely applied due to the drawbacks of cost, difficulty and the time consumed to design models.

For a mine site without any similar experience of LTCC operation available, the numerical modelling provides a predictive and effective tool to assess the top coal cavability. The theories and applications of numerical methods in rock mechanics have been thoroughly discussed [27,34,58,59]. The following content is to review the cavability evaluations that were developed based on various numerical modelling techniques.

3.2.1. Continuum methods

In continuum methods, the problem domain is discretised into many sub-domains with an assumption that the connectivity between the sub-domains does not change as simulation progresses [60]. The Finite Difference Method (FDM), Finite Element Method (FEM) and Boundary Element Method (BEM) are known as the most applied continuum methods in rock mechanics. It is noted that unlike FDM and FEM, BEM requires the discretisation at the boundary of problem domain only. It reduces the problem dimensions and therefore can significantly reduce the computing time. However, BEM is not as efficient as FEM and FDM in dealing with non-homogeneity and non-linearity materials [59]. This probably restricts the application of BEM in longwall studies where stratified coal measure rocks show the high level of heterogeneity and plasticity behaviour.

Finite Difference Method is a direct approximation of the governing partial differential equations by replacing partial derivatives with differences at grids in problem domains [59]. For conventional longwall study, FLAC and FLAC3D codes based on FDM have

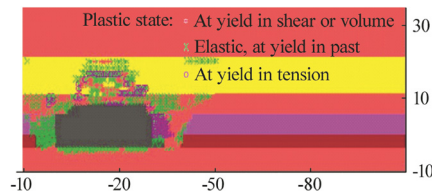


Fig. 6. Longwall progressive failure and caving in FLAC model (after Singh and Singh [63]).

been widely used. The failure characteristics of roof rocks have been studied using FLAC [61,62]. Singh and Singh employed FLAC to study the progressive caving of roof strata (Fig. 6) [63]. A caving criterion was developed based on element shear strain. Shabani-mashcool and Li, on the contrary, considered that rock is more often subjected to tensile failure when caving occurs [64]. Consequently, a longwall mining simulation algorithm using FLAC3D was developed with a caving criterion based on element tensile strain.

For LTCC study, FLAC was used to conduct parametric evaluations of top coal thickness, roof condition and support setting pressure [1]. The study introduced an approach for studying the top coal cavability under different mining conditions. The stress distribution, top coal deformation and fracturing pattern induced by LTCC mining have been investigated in three-dimensional problem using FLAC3D [6,14,15]. The complex mechanisms involved in mining have been successfully investigated using numerical method, which is too difficult to be solved analytically.

Similar to FDM, FEM requires a problem domain to be divided into an assembly of discrete and interacting elements. The state of total stress and induced displacements are determined throughout the global stiffness matrix [27]. The two-dimensional FEM was applied to study stress distribution and failure zones induced above conventional longwall panels [31,46]. Tang developed a numerical code called RPPA which is based on the linear finite element method [65]. The code is able to study the evolution of damage due to progressive failure leading to collapse in brittle rock. The RPPA code has been applied to characterise the development of fracture zones induced by LTCC [18].

The Commonwealth Scientific and Industrial Research Organization (CSIRO) developed an in-house continuum code called COSFLOW. The code is based on FEM and incorporates the layered Cosserat continuum formulation. It is capable of efficiently simulating the rock breakage, slip and separation along bedding planes in weak layered rocks. The COSFLOW code was utilised to assess the top coal cavability [24]. The study suggested that mining depth, coal strength and top coal thickness are the most important factors controlling top coal caving. The multivariate linear regression of parametric studies proposed a relationship between parameters and caving index, as given in Eq. (2):

$$CI = -2.64 + 0.0395H - 0.72CS + 0.191TC \quad (2)$$

where CI is the caving index; H is the mining depth, m; CS is the uniaxial coal strength, MPa; TC is the top coal thickness, m. The caving index was calculated for a range of Chinese LTCC faces. Then, the results were compared with the recoded top coal recovery at real faces. Finally, a cavability classification was developed based on the caving index and top coal recovery values, as shown in Table 8.

This code has also been used to study the top coal failure, strata caving mechanism and stress redistribution in a mine-scale LTCC mining [66].

The continuum methods, in general, provide a useful modelling technique which is capable of analysing the complex caving mechanisms induced by longwall mining. Compared to discontinuum methods, continuum methods are likely to be more efficient regarding computational solution time for large-scale problems. However, continuum methods are limited in their ability to represent large-scale opening, sliding and complete detachment of elements [59]. This may reduce the accuracy of cavability assessments in jointed coal seam.

3.2.2. Discontinuum methods

The most representative discontinuum method for rock mechanics is the Distinct Element Method (DEM). In DEM, a problem domain is represented as an assemblage of blocks formed by connected fractures. The method is most suitable for moderately fractured rock mass where the number of fractures is too large for continuum methods or where the large-scale displacement of block is required [59]. The DEM method has been increasingly used to study longwall caving mechanisms with the commonly applied codes such as UDEC, PFC2D, 3DEC and PFC3D.

29311601009650 For conventional longwall study, UDEC has been used to analyse the roof rock fracturing process, strata movement and rupture mode [67,68]. For a LTCC study, UDEC and 3DEC have been used to evaluate the importance of factors affecting top coal cavability. An evaluation of the UDEC applicability in characterising LTCC caving performance was conducted by CSIRO. The modelling results appear qualitatively plausible with respect to caving span length, peak abutment stress distribution and chock support pressure [69]. A cavability rating system was developed by Vakili and Hebblewhite using UDEC and 3DEC [16]. The approach is similar to that used by Humphries et al. [24]. The first step was to use DEM codes to quantify the impact of parameters on top coal cavability. The second step was to develop relationships between key factors and proposed cavability indicators using statistical analysis, as given in Eqs. (3) and (4).

$$MCD = 2.451E + 0.46T + 12.56J_h + 19.56J_v + 22.417e^{-0.0316\sigma_h} - 23.84 \quad (3)$$

$$TCR = 63.13T^{-0.1514} - 8.85E - 82.447J_v + 0.015\sigma_v^2 + 0.0000401\sigma_v + 91.093 \quad (4)$$

where MCD is the main caving distance, m; TCR is the top coal recovery, %; E is the elastic modulus of intact coal, GPa; T is the top coal thickness, m; J_v is the spacing of sub-vertical joints, m; J_h is the spacing of sub-horizontal joints, m; σ_h is the major horizontal principle stress magnitude, MPa; and σ_v is the vertical principle stress magnitude, MPa. In the last step, a cavability classification called Top Coal Caving Rating (TCCR) was developed using back analysis from past Chinese LTCC practices, as illustrated in Table 9 and Fig. 7.

A different cavability classification for inclined thick coal seams was developed by Dao using 3DEC [11]. The methodology is also similar to that undertaken by Humphries et al. and Vakili and Heb-

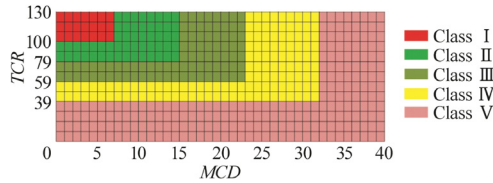
Table 8
Cavability classification in CSIRO's method (after Humphries et al. [24]).

Category	Excellent	Good	Moderate	Poor	Very Poor
Top coal recovery (%)	>80	65–79	50–64	30–49	<30
Caving index CI	>0.70	–4.80 to 0.70	–10.30 to –4.80	–17.64 to –10.30	<–17.64

Table 9

Cavability classification in TCCR system (after Vakili and Hebblewhite [16]).

Class	MCD (m)	TCR (%)
I-extremely weak roof	<7	>100
II-good cavability	7–15	79–100
III-fair cavability	15–23	59–79
IV-poor cavability	23–32	39–59
V-extremely poor cavability	>32	<39

**Fig. 7.** Cavability graph in TCCR system (after Vakili and Hebblewhite [16]).

blewhite [16,24]. The effect of seam dip on cavability was incorporated, as given in Eq. (5).

$$CPI = 73.38 - 3.085E - 0.095\alpha + 48.21T^{-0.2291} - 29.75S_v - 132.17S_h + 0.0336H \quad (5)$$

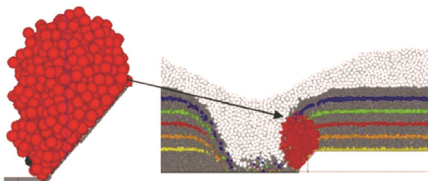
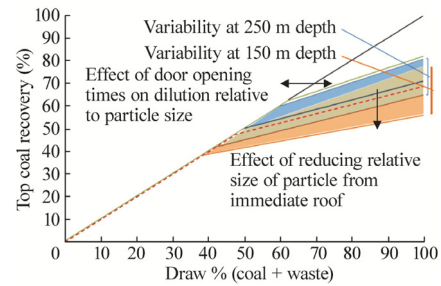
where CPI is the caving performance index; E is the Young's modulus of coal seam, MPa; α is the coal seam dip, degree; T is the top coal thickness, m; S_v is the spacing of vertical joints, m; S_h is the spacing of bedding planes, m; H is the cover depth of coal seam, m. The classification is shown in Table 10.

The UDEC and 3DEC codes have been successfully utilised to simulate the explicit LTCC caving. The effect of top coal discontinuities on caving has been studied in detail. However, these studies are limited in modelling the rock and coal blocks by elastic materials. This can be explained that the interaction of block failure with joint failure may result in extremely complex material response [69]. The elastic materials cannot simulate the failure of intact blocks and thus, reduce the reliability of modelling outputs.

Since the caving of top coal shows some characteristics similar to that of granular material flow, the particle flow codes such as PFC and PFC3D have been employed to study caving mechanisms. With particle flow codes, a material is represented as assemblages of circular or spherical particles. The applicability of PFC in studying LTCC was investigated by CSIRO [70]. The drawing mechanism of broken top coal has been evaluated using PFC and PFC3D [13,17], as shown in Fig. 8. It is noted that particle flow codes are currently

Table 10Cavability classification based on CPI values (after Dao [11]).

Category	Classification	CPI (%)
1	Excellent	>75
2	Good	50–75
3	Medium	25–49
4	Poor	<25

**Fig. 8.** Final status of top coal drawing in PFC3D model (after Wang et al. [17]).**Fig. 9.** Top coal recovery profile (after Medhurst et al. [75]).

restricted to simulating caving in small-scale mining area due to the high computational efficiency required.

An alternative representation of particle model can be conducted using UDEC with Voronoi tessellation [71]. A rock material is represented as an assembly of polygonal blocks bonded together via the contacts between them. The theory and application of UDEC with Voronoi tessellation can be found in Kazerani and Zhao, and Lan et al. [72,73]. For conventional longwall study, this technique has been employed to capture the progressive caving of roof strata [74]. The combination of UDEC with Voronoi tessellation seems to be a promising modelling tool for coal caving study where the top coal fracturing greatly contributes to the success of its caving. However, effort should be made to adequately represent rock mass properties from contact micro-properties and block properties.

Although DEM is an attractive tool to study the top coal cavability, there are two major concerns that must be considered prior to the application. Firstly, the persistence and orientation of discontinuities are generally difficult to obtain from practice. Secondly, the damping mechanism which is necessary to remove elastic strain in DEM should be adequately identified. This affects the level of element displacements and hence, influences the level of cavability.

3.2.3. Hybrid methods

Each numerical method can be efficiently combined with other numerical methods [59]. The combinations are able to model a large-scale mining area with sufficient detail in the area of interest. A hybrid FLAC/PFC method was developed to assess the top coal caving [75]. The near field was modelled using PFC to capture the complex failure mechanisms while the outer fields were modelled using FLAC to represent the impacts of cover depth, overlying strata and abutment stress. The study suggested a top coal recovery profile that indicates the cavability under effects of the articulated rear canopy and immediate roof in 150–250 m cover depths (Fig. 9). In general, the hybrid methods require an intensive consideration on the modelling process such as the continuity at different region boundaries, the mechanical coupling scheme and the communication between codes [71].

4. Discussion and conclusion

There are many factors influencing the applicability of LTCC including ventilation, spontaneous combustion, development system, cultural acceptance and geotechnical issues [1]. The successful application of LTCC, in terms of geotechnical issues, is considered to be highly dependent on the cavability of top coal. Numerous studies have been conducted to evaluate and predict the top coal cavability. However, it is currently recognised that there is no single cavability assessment method that can be universally applied. In an effort to assist the development of a better assessment, the paper has systematically reviewed a number of cavability evaluations with focus on the affecting parameters and the assessment approaches.

A profound understanding of the parameters affecting top coal caving is a prerequisite for the development of a caving evaluation. Through literature, the key geological and geotechnical parameters were identified such as coal strength, coal discontinuities, coal seam thickness, immediate and main roof characteristics and pre-mining stresses. The persistence and spatial orientation of discontinuities, the three-dimensional nature of stress and the characteristics of overlying strata are some examples of factors whose effects on top coal cavability have not been investigated in detail. Further parametric evaluations of these factors are required to achieve a better assessment.

A review of the typical classifications for rock mass and coal measure rocks have highlighted the need for a different classification for the unique characteristics of top coal caving. Empirical, observational, physical modelling and numerical modelling methods have been commonly applied to evaluate top coal caving. Empirical, observational and physical modelling methods appear to be limited in their application as a predictive tool for cavability assessment in various geological mining conditions. In contrast, the numerical modelling methods provide a prominent tool for the study of the complex caving mechanisms induced by LTCC in multiple possibilities of site conditions.

To obtain the success in numerical analysis, researchers should be clearly aware of the capabilities, benefits and limitations of the methods chosen for study. Modellers need to make appropriate assumption, simplification, calibration and validation for modelling to obtain reliable outputs. For a better cavability analysis, the material constitutive model should be able to simulate the rock strength reduction after failure during caving. The goaf material compaction that affects the overlying strata deformation in periodic caving must be taken into consideration. In addition, an explicit representation of the LTCC support is necessary to capture the realistic interaction between support, top coal and surrounding rocks. It is noted that most of LTCC simulations studied the cavability in a limited area of mining. In these models, the roof rock strata were likely to remain undisturbed. This may underestimate the influence of overlying strata on the top coal behaviour. In addition, many LTCC models were two-dimensional that may not incorporate the three-dimensional nature of stress distribution. With the rapid development of computer technology, such issues are able to be solved in larger-scale and three-dimensional models. The comprehensive effect of factors on top coal cavability, thus, can be successfully studied.

An appropriate evaluation index contributes to the comprehensibility and practicability of a cavability assessment. The cavability indexes in past studies were either the recovery rate or fracturing or displacement of top coal. For future estimation, a combined index that can represent both the quality and quantity of caving is recommended. Additionally, the available Chinese LTCC practices have provided a useful source for the validation of cavability rating systems. The information from ongoing LTCC applications in other countries such as Australia and Vietnam would be another source to improve the reliability of future classifications.

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